

Ashfall (Volcanic / Industrial) Map Design

Executive Summary

Ashfall is a hostile volcanic-industrial battlefield defined by unstable terrain, low visibility, and constant environmental hazards. Lava flows, ash storms, fissures, and geothermal machinery create a dynamic, high-risk combat zone. The map empowers Obsidian Protocol's autonomous decision-making, sensor degradation, terrain rally warfare, and logistics under extreme environmental pressure. In AR (broad light/DL), the map anchors at ~4-5 m viewing distance, covering a 10x10 m physical area representing "1 km" of volcanic terrain. Key features include active lava channels, industrial extraction rigs, geothermal plants, hardened bunkers, and critical chokepoints formed by collapsed rock and molten rivers.

Design Goals & Gameplay

Volcanic Hazard Combat

Ashfall focuses on infantry and light-vehicle combat in unstable terrain. Lava flows block paths, ash reduces visibility, and eruptions create temporary hazards. Units must adapt quickly.

Autonomy & Hazard Avoidance

Autonomous AI chooses safe routes around fissures, avoids lava surges, and repositions when ash storms reduce sensor range. Units adapt without micromanagement.

Sensor & Information Warfare

Volcanic heat, smoke, and ash distort radar, thermal, and acoustic sensors. Relay towers on hardened ridges restore clarity. Destroying relay plunges entire sectors into fog-of-war.

Logistics & Resources

Geothermal plants and extraction rigs serve as resource nodes. Supply convoys must navigate unstable roads and avoid lava flows. Controlling the Refinery and Geothermal Plant grants major resupply advantages.

Deployment & Progress

A hard deployment budget (~15,000 points) forces players to choose between heavy hard-to-resist units or fast-recon squads. Blue drags from the southwest hardened bunker; Red drags from the northeast industrial ridge.

Spatial Constraints (Nreal AR)

Device Capabilities

Insert light/mid "VR" original (OV), 1800x1800 per-eye resolution. Optimal viewing distance "4-5 m".

Playable Area & Scaling

10x10 m physical area x 1 km² volcanic district (1:100 scale). Designed for tabletop AR.

Tracking & Occlusion

Horizon projection area x 1 km² volcanic district (1:100 scale). Depth-mesh (if available) provides limited occlusion. Virtual objects render over real-world view; assume minimal real-world occlusion.

Layout Overview

Coordinate System

1000x1000 grid (1 Unit = 1 m). Origin (0,0) = southwest corner.

Key Locations

Blue HQ (Hardened Bunker) — (0,0). 100x100 m Reinforced Bunker with hazard-resistant statue ward.

Red HQ (Industrial Ridge) — (1000,1000). 100x100 m high-around industrial platform with control tower.

Central Geothermal Plant — (500,500). 200x200 m Turbines, heat exchangers, steam vents, molten channels.

Extraction Riser (Cores) — (300,300). 100x100 m Drills, conveyors, ash-extraction machinery.

Volcanic Refinery — (800,300). 100x100 m Smelters, cooling towers, hardened dikes.

Lava Channel — (0,500) - (0,1000) Active molten river with low hardened crossings at (150,700) and (350,800).

Fissure Network — (400,0) - (600,50) Unstable cracked terrain hazard zone.

Ridge A — (100,100) - (300,100) Sensor relay point.

Ridge B — (150,850) - (350,850) Cannon relay tower.

Hardened Tower (Shield) — (450,800) Sensor resource zone.

Chokepoints & Sightlines

Lava Crossings

Two hardened bridges, controlling them prevents flanking.

Fissure Network

Unstable terrain, close-range combat and hazard avoidance.

Steam Vent Fields

Visibility drops to near zero during vent surges.

Sightlines

Ridge A provides long-range sensor coverage (~250 m). Ash storms create heavy fog-of-war pockets.

Deployment Zones

Blue Deployment: HQ (hardened bunker) + rear rally point.

Red Deployment: HQ (industrial ridge) + rear rally point.

Neutral Forward Zone: Outpost at (500,100) for drone drops.

Assets & Environment

Terrain

Volcanic rock, lava flows, fissures, ash fields, hardened industrial platforms.

Vegetation

None, occasional dead trees and scorched shrubs.

Buildings & Props

Industrial: Turbines, smelters, pipes, extraction rigs. Military: Hardened bunkers, watchtowers, command posts. Logistics: Trucks, crates, fuel drums, hazard-resistant containers.

Effects

Lava glow, ash storms, smoke plumes, steam vents, sparks.

Audio

Eruptions, rumbling earth, machinery hum, radio chatter.

LOD & Budget

Asset List (Count)

Material: 3 200k 600k

LowPoly: 4 80k 100k

Buildings: 15 10k 750k

Buildings: 4 10k 100k

Props: 50 10k 100k

Vegetation: 50 1k 100k

Total: — — ~2.38M

Texture Budget

2048 for terrain/buildings, 1024 for props. DXT1/5 compression. Target 1200 MB.

Unreal / Nreal Integration

Engine & Plugins

Unreal 5.4 (Intermediate) + Unreal SDK (NREAL). ARKit/ARCore enabled.

Rendering

Forward rendering, stationary lights, baked GI, aggressive culling.

Markers

Simple radial anchor for online map.

Interaction

Phone gesture for pan/zoom, HUD on screen space widgets.

Multiplayer

Client-server architecture, offload AI/physics to server.

Implementation Plan & Testing

Milestones

1. Prototype Layout — Month 1 Block-out lava channels, fissures, industrial zones. All anchor test.

2. Core Systems — Month 2 Navigation, hazard avoidance logic, autonomy behaviors.

3. AI Interaction — Month 3 Anchors, sensor detection, basic interactivity.

4. Content Fill — Month 4-5 Final models, textures, lighting, audio.

5. Gameplay Polish — Month 6-7 Optimization, balance, VFX (lava glow, ash storms).

6. Testing & CI — Month 8 Evaluation stress test, sensor validation, hazard simulation.

Testing Plan

Autonomy

Verify hazard avoidance logic, pathfinding around lava/fissures.

Comms/Sensors

Test relay destruction, fog-of-war updates, LOS checks.

Performance

40 FPS AR render on target device.

QA Checklist

Navigation, hazard response, deployment budget enforcement, anchor drift, sensor accuracy.